

开源软件实践与安全分析

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SHA256:

06c388171d1844556b7baa0d4fd21ee5fb332a56d8

0015a63e0466cad1d44bc

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一、摘要

本报告针对二进制文件 `dhcp6s` 进行了系统的逆向分析。该文件为 ELF 32-bit MIPS 架构可执行文件，动态链接，已去除符号表。通过 `readelf` 分析导入函数，发现包含 `recvmsg`、`inet_ntop` 等网络函数，以及 `dhcp6_get_options`、`transmit_sa`、`is_leased` 等 DHCPv6 协议相关函数。通过 `strings` 提取可读字符串，定位到 `/etc/dhcp6s.conf` 配置文件和 `/var/run/dhcp6s.pid` 文件。使用 `radare2` 对 `main` 函数进行反汇编分析，识别出 20 余次函数调用和 10 个以上条件分支。综合分析确定，`dhcp6s` 是一个运行在 MIPS 架构嵌入式设备上的 DHCPv6 服务端程序。分析过程中使用了 `file`、`readelf`、`strings`、`radare2` 等工具，涵盖了文件识别、导入表分析、字符串提取、反汇编与控制流分析等核心二进制分析技术。

```
wanglingyu@LAPTOP-NC0K492J:/mnt/c/Users/刘憨憨/Desktop/wly/期末大作业$ ls -lh dhcp6s
-rwxrwxrwx 1 root root 175K Jun 13 00:00 dhcp6s
```

二、分析环境

Ubuntu 发行版

Powershell

`wsl --install -d Ubuntu`

创建用户

text

Enter new UNIX username: wanglingyu

New password: ***

Retype password: ***

```
provisioning the new WSL instance ubuntu
This might take a while...
Create a default Unix user account: wanglingyu
New password:
Retype new password:
passwd: password updated successfully
usermod: no changes
Help improve Ubuntu!
```

更新并安装工具

bash

sudo apt update

```
wanglingyu@LAPTOP-NCOK492J:/mnt/c/Windows/system32$ sudo apt update
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute InRelease [136 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [247 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [136 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute/main amd64 Packages [1480 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [64.2 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [31.2 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/main amd64 c-n-f Metadata [3984 B]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [126 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [39.2 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute/main Translation-en [524 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute/main amd64 Components [395 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute/main amd64 c-n-f Metadata [32.4 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute/universe amd64 Packages [16.0 MB]
Get:16 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.1 kB]
Get:17 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 c-n-f Metadata [3324 B]
Get:18 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [219 kB]
Get:19 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [38.6 kB]
Get:20 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 c-n-f Metadata [396 B]
Get:21 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Components [212 B]
Get:22 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 c-n-f Metadata [120 B]
Get:23 http://archive.ubuntu.com/ubuntu resolute/universe Translation-en [6329 kB]
Get:24 http://archive.ubuntu.com/ubuntu resolute-backports/main amd64 Packages [1480 kB]
Get:25 http://archive.ubuntu.com/ubuntu resolute-backports/main amd64 c-n-f Metadata [112 B]
Get:26 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Packages [16.0 MB]
Get:27 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 c-n-f Metadata [116 B]
Get:28 http://archive.ubuntu.com/ubuntu resolute-backports/restricted amd64 Packages [219 kB]
Get:29 http://archive.ubuntu.com/ubuntu resolute-backports/restricted amd64 c-n-f Metadata [120 B]
Get:30 http://archive.ubuntu.com/ubuntu resolute-backports/multiverse amd64 Packages [212 B]
Get:31 http://archive.ubuntu.com/ubuntu resolute-backports/multiverse amd64 c-n-f Metadata [120 B]
Get:32 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [247 kB]
Get:33 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 c-n-f Metadata [64.2 kB]
Get:34 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [31.2 kB]
Get:35 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 c-n-f Metadata [3984 B]
Get:36 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [126 kB]
Get:37 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 c-n-f Metadata [39.2 kB]
Get:38 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [219 kB]
Get:39 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 c-n-f Metadata [38.6 kB]
Get:40 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [212 B]
Get:41 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 c-n-f Metadata [120 B]
Get:42 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [38.6 kB]
Get:43 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 c-n-f Metadata [392 B]
Get:44 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [3328 B]
Get:45 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse Translation-en [772 B]
Get:46 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Components [216 B]
Get:47 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 c-n-f Metadata [256 B]
Get:48 http://archive.ubuntu.com/ubuntu resolute-backports/main amd64 Components [212 B]
Get:49 http://archive.ubuntu.com/ubuntu resolute-backports/main amd64 c-n-f Metadata [112 B]
Get:50 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [216 B]
Get:51 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 c-n-f Metadata [116 B]
Get:52 http://archive.ubuntu.com/ubuntu resolute-backports/restricted amd64 Components [216 B]
Get:53 http://archive.ubuntu.com/ubuntu resolute-backports/restricted amd64 c-n-f Metadata [120 B]
Get:54 http://archive.ubuntu.com/ubuntu resolute-backports/multiverse amd64 Components [216 B]
Get:55 http://archive.ubuntu.com/ubuntu resolute-backports/multiverse amd64 c-n-f Metadata [120 B]
Fetched 32.6 MB in 6s (5090 kB/s)
50 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

bash

sudo apt install -y binutils file gdb radare2 ltrace strace

```
wanglingyu@LAPTOP-NCOK492J:/mnt/c/Windows/system32$ sudo apt install -y binutils file gdb radare2 ltrace strace
binutils is already the newest version (2.46-3ubuntu2).
binutils set to manually installed.
file is already the newest version (1:5.46-5build2).
file set to manually installed.
Installing:
  gdb ltrace radare2 strace
Installing dependencies:
  libbabeltrace1 libb64-dev libdebuginfod-common libipt2 libradare2-6.0.0r64 libsource-highlight-common libuv1-dev libzip-dev manpages-dev zipmerge
  libc-dev-bin libcapstone-dev libdebuginfod1t64 libl24-dev libradare2-common libsource-highlight4t64 libuv1t64 libzip5 rpcsvc-proto ziptool
  libc6-dbg libcapstone5 libdw1t64 libmagic-dev libradare2-dev libunwind9 libxxhash-dev linux-libc-dev zipcmp zlib1g-dev
Suggested packages:
  gdb-doc gdbserver libc-devtools glibc-doc libuv1-doc
Summary:
  Upgrading: 0, Installing: 34, Removing: 0, Not Upgrading: 50
  Download size: 30.1 MB
  Space needed: 116 MB / 1025 GB available
```

```

Setting up libunwind8:amd64 (1.8.3-0ubuntu1) ...
Setting up libsource-highlight-common (3.1.9-4.3build2) ...
Setting up libc6-dbg:amd64 (2.43-2ubuntu2) ...
Setting up libdwlt64:amd64 (0.194-4) ...
Setting up rpcsvc-proto (1.4.3-1build1) ...
Setting up strace (6.19+ds-0ubuntu5) ...
Setting up libipt2 (2.1.2-3) ...
Setting up libcapstone5:amd64 (5.0.7-2) ...
Setting up libbabeltrace1:amd64 (1.5.11-5build1) ...
Setting up libxxhash-dev:amd64 (0.8.3-2build1) ...
Setting up zipcmp (1.11.4-2) ...
Setting up librada2-6.0.0t64:amd64 (6.0.7+ds-1) ...
Setting up libc-dev-bin (2.43-2ubuntu2) ...
Setting up ziptool (1.11.4-2) ...
Setting up libcapstone-dev:amd64 (5.0.7-2) ...
Setting up libuv1-dev:amd64 (1.51.0-2ubuntu1) ...
Setting up libdebuginfod1t64:amd64 (0.194-4) ...
Setting up libsource-highlight4t64:amd64 (3.1.9-4.3build2) ...
Setting up gdb (17.1-2ubuntu1) ...
Setting up liblz4-dev:amd64 (1.10.0-8) ...
Setting up radare2 (6.0.7+ds-1) ...
Setting up libc6-dev:amd64 (2.43-2ubuntu2) ...
Setting up zlib1g-dev:amd64 (1:1.3.dfsg+really1.3.1-1ubuntu3) ...
Setting up libzip-dev:amd64 (1.11.4-2) ...
Setting up librada2-dev (6.0.7+ds-1) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2) ...
ldconfig: /usr/lib/wsl/lib/libcudart.so.1 is not a symbolic link

```

工具安装验证

Bash

file --version

strings --version

gdb --version

```

wanglingyu@LAPTOP-NCOK492J:/mnt/c/Windows/system32$ file --version
--version
gdb --versionfile-5.46
magic file from /etc/magic:/usr/share/misc/magic
wanglingyu@LAPTOP-NCOK492J:/mnt/c/Windows/system32$ strings --version
GNU strings (GNU Binutils for Ubuntu) 2.46
Copyright (C) 2026 Free Software Foundation, Inc.
This program is free software; you may redistribute it under the terms of
the GNU General Public License version 3 or (at your option) any later version.
This program has absolutely no warranty.
wanglingyu@LAPTOP-NCOK492J:/mnt/c/Windows/system32$ gdb --version
GNU gdb (Ubuntu 17.1-2ubuntu1) 17.1
Copyright (C) 2025 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software; you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

```

三、静态分析

3.1 文件完整性验证：sha 结果一致

```
wanglingyu@LAPTOP-NC0K492J:/mnt/c/Users/刘慈恩/Desktop/wly/期末大作业$ sha256sum dhcp6s
06c388171d1844556b7baa0d4fd21ee5fb332a56d80015a63e0466cadcd44bc  dhcp6s 0x469ce8
```

3.2 分析文件

file dhcp6s

```
wanglingyu@LAPTOP-NC0K492J:/mnt/c/Users/刘慈恩/Desktop/wly/期末大作业$ ls -la
total 176
drwxrwxrwx 1 root root 512 Jun 27 13:58
drwxrwxrwx 1 root root 512 Jun 27 13:29
-rwxrwxrwx 1 root root 178656 Jun 13 00:00 dhcp6s
-rwxrwxrwx 1 root root 162 Jun 27 13:49 '$ Microsoft Word '$\346\226\207\346\241\243'.docx'
-rwxrwxrwx 1 root root 0 Jun 27 13:29 '$\346\226\260\345\273\272' Microsoft Word '$\346\226\207\346\241\243'.docx'
dhcp6s: ELF 32-bit LSB executable, MIPS, MIPS-II version 1 (SYSV), dynamically linked, interpreter /lib/ld-uClibc.so.0, stripped
```

信息	含义
ELF 32-bit	32 位 Linux 可执行文件
MIPS	不是 x86/x64 架构，是 MIPS 架构 （嵌入式设备/路由器常用）
dynamically linked	动态链接，依赖外部库
stripped	去除了符号表（函数名被删了，分析难度稍大）

3.3 文件内容识别

```
wanglingyu@LAPTOP-NC0K492J:/mnt/c/Users/刘慈恩/Desktop/wly/期末大作业$ strings dhcp6s | head -80
/lib/ld-uClibc.so.0
_fini
__uClibc_main
__deregister_frame_info
__register_frame_info
__Jv_RegisterClasses
stderr
fwrite
exit
dhcp6_vbuf_free
dhcp6_copy_list
sipnamelist
siplist
dnslist
dnsnamelist
ntplist
nisnamelist
nislist
nisnamelist
nisplist
bcmcsnamelist
bcmcslist
optrefreshtime
dprintf
dhcp6_add_listval
dhcp6_clear_list
duidcmp
dhcp6_find_listval
unlink
dhcp6_set_options
dhcp6_init_options
dhcp6_vbuf_copy
memset
dhcp6_clear_options
outsock
transmit_sa
dhcp6msgstr
dhcp6_calc_mac
```

```
find_pool
is_available_in_pool
get_free_address_from_pool
device
duidfree
release_address
duidstr
dhcp6_set_timer
dhcp6_clear_listval
is_leased
malloc
duidcpy
dhcp6_add_timer
in6addr2str
dhcp6_remove_timer
sprintf
system
dhcp6_verify_mac
memcpy
cfparse
__errno_location
find_hostconf
get_rdtype
dhcp6_auth_replaycheck
memcmp
dhcp6_validate_key
access
decline_address
create_dynamic_hostconf
dhcp6_check_timer
insock
ctlsock
dhcp6_ctl_setreadfds
select
dhcp6_ctl_readcommand
recvmsg
__cmsg_nxthdr
ifidx
find_ifconfbyid
dhcp6_ctl_acceptcommand
dhcp6_get_options
cause
```

3.4 readelf 导入函数分析

```
readelf -s dhcp6s | grep FUNC | head -30
```

```
wanglingyu@LAPTOP-NCOK492J:/mnt/c/Users/刘慈恩/Desktop/wly/期末大作业$ readelf -s dhcp6s | grep FUNC | head -30
12: 004207a4 96 FUNC GLOBAL DEFAULT 8 yyerror
15: 00422190 572 FUNC GLOBAL DEFAULT UND free
20: 00422180 84 FUNC GLOBAL DEFAULT UND close
21: 00000000 0 FUNC WEAK DEFAULT UND __register_frame_info
23: 00420a24 380 FUNC GLOBAL DEFAULT 8 cfparse
25: 00422170 84 FUNC GLOBAL DEFAULT UND recvmsg
26: 00409a20 76 FUNC GLOBAL DEFAULT 8 dhcp6_vbuf_free
27: 00000000 0 FUNC WEAK DEFAULT UND _Jv_RegisterClasses
29: 0040a65c 128 FUNC GLOBAL DEFAULT 8 transmit_sa
30: 00422160 148 FUNC GLOBAL DEFAULT UND fputs
31: 00422150 44 FUNC GLOBAL DEFAULT UND strchr
32: 00414f78 2032 FUNC GLOBAL DEFAULT 8 configure_host
33: 00413070 580 FUNC GLOBAL DEFAULT 8 configure_global[...]
34: 0041e4f0 8608 FUNC GLOBAL DEFAULT 8 yylex
35: 00422140 112 FUNC GLOBAL DEFAULT UND open
37: 00422130 36 FUNC GLOBAL DEFAULT UND strlen
41: 0040af60 356 FUNC GLOBAL DEFAULT 8 duidstr
42: 00409f70 124 FUNC GLOBAL DEFAULT 8 dhcp6_vbuf_copy
43: 00422120 140 FUNC GLOBAL DEFAULT UND fileno
45: 00422110 28 FUNC GLOBAL DEFAULT UND atoi
46: 0040d950 1120 FUNC GLOBAL DEFAULT 8 get_duid
47: 0040f8a0 4424 FUNC GLOBAL DEFAULT 8 dhcp6_get_options
48: 0040b65c 172 FUNC GLOBAL DEFAULT 8 setloglevel
50: 00422100 864 FUNC GLOBAL DEFAULT UND inet_ntop
51: 00417b94 68 FUNC GLOBAL DEFAULT 8 is_leased
52: 00417584 252 FUNC GLOBAL DEFAULT 8 dhcp6_set_timer
53: 004220f0 192 FUNC GLOBAL DEFAULT UND exit
54: 004220e0 20 FUNC GLOBAL DEFAULT UND __errno_location
55: 0041def8 160 FUNC GLOBAL DEFAULT 8 yy_init_buffer
56: 00413810 1452 FUNC GLOBAL DEFAULT 8 configure_commit
```

3.4.1 网络通信相关函数（核心功能）

函数名	作用
recvmsg	接收网络数据包
inet_ntop	将 IP 地址转换为字符串格式

结论：这个程序具有网络通信能力，能够接收和发送网络数据包。

3.4.2 DHCP 相关函数（关键线索）

函数名	作用
dhcp6_get_options	解析 DHCPv6 选项
dhcp6_set_timer	设置 DHCP 定时器
dhcp6_vbuf_*	DHCPv6 缓冲区管理

函数名	作用
is_leased	检查地址是否已租用
get_duid	获取 DHCP 唯一标识符
configure_host	配置主机
configure_global	配置全局参数
configure_commit	提交配置

结论：这是一个 **DHCPv6 服务器** 程序，运行在 MIPS 架构设备（如路由器）上。

3.4.3 文件操作相关函数

函数名	作用
open	打开文件
free	释放内存
close	关闭文件句柄
fileno	获取文件描述符

结论：程序会读写配置文件或日志文件。

3.4.4 其他

函数名	作用
yyerror	语法错误处理

函数名	作用
yylex	词法分析器
yy_init_buffer	初始化解析缓冲区
fputs	输出字符串到文件
strchr	字符串查找
strlen	获取字符串长度
atoi	字符串转整数
exit	程序退出

结论：程序包含**配置文件解析功能**（用 Flex/Bison 生成的解析器），会读取配置文件。

3.5 字符串分析

3.5.1 查看字符串找配置文件路径

Bash

```
strings dhcp6s | grep -E "\.(conf|pid|log|lease)"
```

```
wanglingyu@LAPTOP-NC0K492J:/mnt/c/Users/刘慈恩/Desktop/wly/期末大作业$ strings dhcp6s | grep -E "\.(conf|pid|log|lease)"
/var/run/dhcp6s.pid
/etc/dhcp6s.conf
```

通过 `strings dhcp6s | grep -E "\.(conf|pid|log|lease)"` 筛选出配置相关路径：

字符串	推断用途
/etc/dhcp6s.conf	程序启动时读取的配置文件
/var/run/dhcp6s.pid	守护进程记录 PID 的文件

结论：存在 .conf 配置文件，结合导入函数中的 yyparse/yylex（Flex/Bison 生成的解析

器)，证实该程序依赖外部配置文件运行。

3.6 反汇编分析

3.6.1 用 radare2 找 main 函数入口

Bash

r2 -A dhcp6s

```
wanglingyu@LAPTOP-NCOK492J: /mnt/c/Users/刘慈恩/Desktop/wly/期末大作业$ r2 -A dhcp6s
WARN: Relocs has not been applied. Please use '-e bin.relocs.apply=true' or '-e bin.cache=true' next time
INFO: Analyze all flags starting with sym. and entry0 (aa)
INFO: Analyze imports (af@@@i)
INFO: Analyze entrypoint (af@ entry0)
INFO: Analyze symbols (af@@@s)
INFO: Analyze all functions arguments/locals (afva@@F)
INFO: Analyze function calls (aac)
INFO: Analyze len bytes of instructions for references (aar)
INFO: Finding and parsing C++ vtables (avrr)
INFO: Analyzing methods (af @@ method.*)
INFO: Emulate functions to find computed references (aaef)
INFO: Function is too sparse, must be analyzed with recursive
INFO: Recovering local variables (afva@@F)
INFO: Type matching analysis for all functions (aافت)
INFO: Propagate noreturn information (aanr)
INFO: Use -AA or aaaa to perform additional experimental analysis
INFO: Finding xrefs in noncode sections (e anal.in=io.maps.x; aav)
[0x004026d0] > pdf @ main
Do you want to print 2285 lines? (y/N) y
; NULL XREF from section .dynsym @ 0x654(r)
; DATA XREF from entry0 @ 0x4026ec(r)
3800: int main (socklen_t optlen, int argc, char **argv, char **envp, int32_t arg4, int32_t arg_0h, int32_t arg_4h);
- args(zero, a0, a1, a2, a3, sp[0x0..0x4]) vars(32:sp[0x4..0x90])
0x00408eb4 07001c3c lui gp, 7
0x00408eb8 cc89c27 addiu gp, gp, -0x7634
0x00408ebc 21e09903 addu gp, gp, t9
0x00408ec0 60f1bd27 addiu sp, sp, -0xa0
0x00408ec4 9c00bfaf sw ra, (var_9ch)
0x00408ec8 9800beaf sw fp, (var_98h)
0x00408ecc 9400b7af sw s7, (var_94h)
0x00408ed0 9000b6af sw s6, (var_90h)
0x00408ed4 8c00b5af sw s5, (var_8ch)
0x00408ed8 8800b4af sw s4, (var_88h)
0x00408edc 8400b3af sw s3, (var_84h)
0x00408ee0 8000b2af sw s2, (var_80h)
0x00408ee4 7c00b1af sw s1, (var_7ch)
0x00408ee8 7800b0af sw s0, (var_78h)
0x00408eec 1800bcdf sw gp, (var_18h)
0x00408ef0 0000b08c lw s0, (a1) ; argv
0x00408ef4 cc82998f lw t9, -sym.imp.strrchr(gp) ; [0x421e40:4]=0x8f998010
0x00408ef8 2188a000 move s1, a1
0x00408efc 21988000 move s3, a0
0x00408f00 2f000524 addiu a1, zero, 0x2f ; argv
0x00408f04 09f82003 jair t9
0x00408f08 21200002 move a0, s0
```

```
0x00409ef4 2080998f lw t9, -segment.LOAD0(gp) ; [0x400000:4]=0x404c457f ; segment.ehdr
0x00409ef8 00000000 nop
0x00409efc 08723927 addiu t9, t9, 0x7208 ; 0x407208 ; fcn.00407208
0x00409f00 09f82003 jair t9
0x00409f04 00000000 nop
0x00409f08 1800bc8f lw gp, (var_18h)
0x00409f0c 00000000 nop
0x00409f10 dc80998f lw t9, -sym.imp.exit(gp) ; [0x4220f0:4]=0x8f998010
0x00409f14 00000000 nop
0x00409f18 09f82003 jair t9
0x00409f1c 21200000 move a0, zero
0x00409f20 CODE XREF from 0x4099e0(x)
0x00409f24 e080998f lw t9, -sym.imp.__errno_location(gp) ; [0x4220e0:4]=0x8f998010
0x00409f28 00000000 nop
0x00409f2c 09f82003 jair t9
0x00409f30 00000000 nop
0x00409f34 1800bc8f lw gp, (var_18h) ; [0x4698cb4:4]=0
0x00409f38 0000448c lw a0, (v0) ; obj.outsock
0x00409f3c e883998f lw t9, -sym.imp.strerror(gp) ; [0x421c90:4]=0x8f998010
0x00409f40 00000000 nop
0x00409f44 09f82003 jair t9
0x00409f48 00000000 nop
0x00409f4c 1800bc8f lw gp, (var_18h)
0x00409f50 21384000 move a3, v0
0x00409f54 00000000 lw a1, -0x7fd(gp) ; [0x4698a4:4]=0x420000
0x00409f58 2480868f lw a2, -0x7fd(gp) ; [0x4698a4:4]=0x420000
0x00409f5c 4c82998f lw t9, -sym.dprintf(gp) ; [0x40b708:4]=0x3c1c0006
0x00409f60 9c38a524 addiu a1, a1, 0x389e ; 0x42389e ; "server6_init" ; argv ; str.server6_init
0x00409f64 3df00010 b 0x409a58 ; 0x4235d0 ; "socket(outsock): %s" ; envp ; str.socket_outsock ; __s
0x00409f68 d035c624 addiu a2, a2, 0x35d0 ; 0x4235d0 ; "socket(outsock): %s" ; envp ; str.socket_outsock ; __s
0x00409f6c CODE XREF from 0x40991d4(x)
0x00409f70 2480868f lw a1, -0x7fd(gp) ; [0x4698a4:4]=0x420000
0x00409f74 2480868f lw a2, -0x7fd(gp) ; [0x4698a4:4]=0x420000
0x00409f78 4c82998f lw t9, -sym.dprintf(gp) ; [0x40b708:4]=0x3c1c0006
0x00409f7c 9c38a524 addiu a1, a1, 0x3894 ; 0x423894 ; "main" ; argv ; str.main
0x00409f80 00000000 nop
0x00409f84 04ff0010 b 0x409998
0x00409f88 00000000 nop
envp ; str.do_not_specify_DNS_servers_both_by_command_line_and_by_co
nfiguration file.
0x00409f78 e435c624 addiu a2, a2, 0x35e4 ; 0x4235e4 ; "do not specify DNS servers both by command line and by configuration file."
0x00409f7c 09f82003 jair t9
0x00409f80 00000424 addiu a0, zero, 6 ; argc
0x00409f84 04ff0010 b 0x409998
0x00409f88 00000000 nop
```

3.6.2 main 函数反汇编

在 radare2 中执行:

Bash

pdf @ main | head -100

pdf @ main > main_disasm.txt

```
[0x004026d0]> # 只显示main函数的前100行 (更清晰)
[0x004026d0]> pdf @ main | head -100
出到文件保存
pdf @ main > main_disasm.txt
; NULL XREF from section..dynsym @ +0x654(r)
; DATA XREF from entry0 @ 0x4026ec(r)
-3800: int main (socklen_t optlen, int argc, char **argv, char **envp, int32_t arg4, int32_t arg_0h, int32_t arg_4h):
- args(zero, a0, a1, a2, a3, sp[0x0..0x4]) vars(32:sp[0x4..0x90])
0x00408eb4 07001c3c lui gp, 7
0x00408eb8 cc899c27 addiu gp, gp, -0x7634
0x00408ebc 21e09903 addu gp, gp, t9
0x00408ec0 60ffbd27 addiu sp, sp, -0xa0
0x00408ec4 9c00bfaf sw ra, (var_9ch)
0x00408ec8 9800beaf sw fp, (var_98h)
0x00408ecc 9400b7af sw s7, (var_94h)
0x00408ed0 9000b6af sw s6, (var_90h)
0x00408ed4 8c00b5af sw s5, (var_8ch)
0x00408ed8 8800b4af sw s4, (var_88h)
0x00408edc 8400b3af sw s3, (var_84h)
0x00408ee0 8000b2af sw s2, (var_80h)
0x00408ee4 7c00b1af sw s1, (var_7ch)
0x00408ee8 7800b0af sw s0, (var_78h)
0x00408eec 1800beaf sw gp, (var_18h)
0x00408ef0 0000b08c lw s0, (a1)
0x00408ef4 cc82998f lw t9, -sym.imp.strchr(gp) ; argv
0x00408ef8 2188a000 move s1, a1
0x00408efc 21988000 move s3, a0
0x00408f00 2f000524 addiu a1, zero, 0x2f ; argv
```

```
[0x004026d0]> pdf @ main > main_disasm.txt
[0x004026d0]> cat main_disasm.txt
; NULL XREF from section..dynsym @ +0x654(r)
; DATA XREF from entry0 @ 0x4026ec(r)
-3800: int main (socklen_t optlen, int argc, char **argv, char **envp, int32_t arg4, int32_t arg_0h, int32_t arg_4h):
- args(zero, a0, a1, a2, a3, sp[0x0..0x4]) vars(32:sp[0x4..0x90])
0x00408eb4 07001c3c lui gp, 7
0x00408eb8 cc899c27 addiu gp, gp, -0x7634
0x00408ebc 21e09903 addu gp, gp, t9
0x00408ec0 60ffbd27 addiu sp, sp, -0xa0
0x00408ec4 9c00bfaf sw ra, (var_9ch)
0x00408ec8 9800beaf sw fp, (var_98h)
0x00408ecc 9400b7af sw s7, (var_94h)
0x00408ed0 9000b6af sw s6, (var_90h)
0x00408ed4 8c00b5af sw s5, (var_8ch)
0x00408ed8 8800b4af sw s4, (var_88h)
0x00408edc 8400b3af sw s3, (var_84h)
0x00408ee0 8000b2af sw s2, (var_80h)
0x00408ee4 7c00b1af sw s1, (var_7ch)
```

3.6.3 图形化模式

The screenshot shows a disassembler interface with the following content:

- Disassembly:**

```

0x004026b0 .string ".in6addr_loopback": len=19
0x004026b1 .string ".fext": len=7
0x004026b2 .string ".fdat": len=7
0x004026b3 .string ".sp": len=4
0x004026b4 .string ".edata": len=7
0x004026b5 .string ".bss_start": len=12
0x004026b6 .string ".fbss": len=6
0x004026b7 .string ".end": len=5
0x004026b8 00
0x004026b9 .section .init:
0x004026ba .NULL IEEE from section, dynamic 0x004026ba
0x004026bb .NULL IEEE from section, dynamic 0x004026bb
0x004026bc .DATA IEEE from entry 0 & 0x004026bc
136: sym_init(0)
afv: vars(3:sp(0x4),0x10)
0x004026bd 0701c3c lui gp, 7
0x004026be 4029c27 addiu gp, gp, -0x004026be
0x004026bf 21e9993 addiu gp, gp, t9
0x004026c0 e07b427 addiu gp, gp, -0x20
0x004026c1 107bcaf sw gp, (var_10h)
0x004026c2 1c0bcaf sw ra, (var_1ch)
0x004026c3 187bcaf sw gp, (var_18h)
0x004026c4 0101104 bal 0x004026c4
0x004026c5 0000000 nop
0x004026c6 .CALL IEEE from sym_init @ 0x004026c6
0x004026c7 0701c3c lui gp, 7
0x004026c8 1c129c27 addiu gp, gp, -0x004026c8
0x004026c9 21e9993 addiu gp, gp, ra
0x004026ca 208999f lw t9, -segment.LOAD0(gp)
0x004026cb 0000000 nop
0x004026cc 00283927 addiu t9, t9, 0x2800
0x004026cd 09f82003 jalr t9
0x004026ce 0000000 nop
0x004026cf 107bcaf lw gp, (var_10h)
0x004026d0 0000000 nop
0x004026d1 0101104 bal 0x004026d1
0x004026d2 0000000 nop
0x004026d3 .CALL IEEE from sym_init @ 0x004026d3
0x004026d4 0701c3c lui gp, 7
0x004026d5 ecf19c27 addiu gp, gp, -0x004026d5
0x004026d6 21e9993 addiu gp, gp, ra
0x004026d7 248999f lw t9, -0x7f0(gp)
0x004026d8 0000000 nop
0x004026d9 a01b3927 addiu t9, t9, 0x1ba0

```
- Functions:**

```

0x00422190 1 32 sym.imp.free
0x00422190 1 16 sym.imp.close
0x00422170 1 16 sym.imp.recvmsg
0x00422160 1 16 sym.imp.puts
0x00422150 1 16 sym.imp.strchr
0x00422140 1 16 sym.imp.open
0x00422130 1 16 sym.imp.strlen
0x00422120 1 16 sym.imp.fileno
0x00422110 1 16 sym.imp.atol
0x00422100 1 16 sym.imp.inet_ntop
0x004220f0 1 16 sym.imp.exit
0x004220e0 1 16 sym.imp._errno_location
0x004220d0 1 16 sym.imp.strptime
0x004220c0 1 16 sym.imp.access
0x004220b0 1 16 sym.imp.fwrite
0x004220a0 1 16 sym.imp.getaddrinfo
0x00422090 1 16 sym.imp.random
0x00422080 1 16 sym.imp.setlogmask
0x00422070 1 16 sym.imp.sprintf
0x00422060 1 16 sym.imp.fsetc
0x00422050 1 16 sym.imp.strcmp
0x00422040 1 16 sym.imp.syslog

```
- Symbols:**

```

0x004697a0 0 fd_data
0x00400001 0 _DYNAMIC_LINKING
0x00471880 0 _gp
0x004026a0 0 ftext
0x00469880 0 _RLD_MAP
0x00469cb0 0 _bss_start
0x00469cb0 0 _edata
0x00469890 0 _GLOBAL_OFFSET_TABLE_
0x00464e30 0 end
0x00469cb0 0 fbss
0x004201e4 96 yerror
0x00469864 4 lineno
0x00469ccc 4 debug_thresh
0x00469838 4 cf_sip_name_list
0x00469d3c 8 siplist
0x00469d34 8 siplist
0x00469d84 4 yychar
0x00420324 350 cfpars
0x00469d44 4 timer_head
0x00469e20 76 dhcp6_ybuf_free
0x00469d94 4 configfilename
0x00404b5c 128 transmit_sa

```

3.6.4 反汇编分析 - 关键函数调用

The screenshot shows a command prompt with the following output:

```

[0x004026d0]> / recvmmsg
0x00401e96 hit0_0 .ctl_readcommandrecvmmsg__cmsg_nxthdri.
0x00423058 hit0_1 .t packet info:recvmmsg: %sreceived %s.

```

结论：程序通过 recvmmsg 接收网络数据包，并有日志记录功能。

```

[0x004026d0]> / dhcp6
0x00401b4e hitl_0 .errfwriteexitdhcp6_vbuf_freedhcp6.
0x00401b5e hitl_1 .dhcp6_vbuf_freedhcp6_copy_listsipna.
0x00401bf7 hitl_2 .eshtimedprintfdhcp6_add_listvaldhc.
0x00401c09 hitl_3 .cp6_add_listvaldhcp6_clear_listduid.
0x00401c22 hitl_4 .ar_listduidcmpdhcp6_find_listvalun.
0x00401c3c hitl_5 .listvalunlinkdhcp6_set_optionsdhc.
0x00401c4e hitl_6 .cp6_set_optionsdhcp6_init_optionsdh.
0x00401c61 hitl_7 .p6_init_optionsdhcp6_vbuf_copymemse.
0x00401c78 hitl_8 .buf_copymemsetdhcp6_clear_optionso.
0x00401ca0 hitl_9 .ocktransmit_sadhcp6msgstrdhcp6_cal.
0x00401cac hitl_10 ._sadhcp6msgstrdhcp6_calc_macfind_p.
0x00401d1d hitl_11 .addressduidstrdhcp6_set_timerdhcp6.
0x00401d2d hitl_12 .dhcp6_set_timerdhcp6_clear_listvali.
0x00401d5a hitl_13 .mallocduidcpydhcp6_add_timerin6ad.
0x00401d76 hitl_14 .merin6addr2strdhcp6_remove_timersp.
0x00401d98 hitl_15 .sprintfsystemdhcp6_verify_macmemc.
0x00401de3 hitl_16 .onfget_rdvaledhcp6_auth_replaychec.
0x00401e01 hitl_17 .laycheckmemcmpdhcp6_validate_keyac.
0x00401e43 hitl_18 .ynamic_hostconfdhcp6_check_timerins.
0x00401e64 hitl_19 .insockctlsockdhcp6_ctl_setreadfds.
0x00401e80 hitl_20 .treadfdselectdhcp6_ctl_readcommand.
0x00401ec2 hitl_21 .find_ifconfbyiddhcp6_ctl_acceptcomma.
0x00401eda hitl_22 .l_acceptcommanddhcp6_get_optionscau.
0x00401f62 hitl_23 .fprintffclosedhcp6_move_listlease.
0x00401f95 hitl_24 .oindexget_duiddhcp6_ctl_authinitge.
0x0040200b hitl_25 .onerrxctladdrdhcp6_ctl_initopenlo.
0x0040202f hitl_26 .oggai_strerrordhcp6_vbuf_cmpsendto.
0x004020b2 hitl_27 .gclock_gettimedhcp6_event_statestr.
0x004020d2 hitl_28 .estrsol_max_rtdhcp6optstrstrlen_.
0x00402141 hitl_29 .6_addrscopebyifdhcp6_remove_evdatas.
0x0040216a hitl_30 .atprefix6_maskdhcp6_stcodestrputs.
0x004021e4 hitl_31 .ushstrchratoldhcp6_iffind_ifconfb.
0x00402251 hitl_32 .64_decodestringdhcp6_modecf_sip_lis.
0x0040239a hitl_33 .nferroracceptdhcp6_ctl_closecomman.
0x00422225 hitl_34 .:1/var/run/dhcp6s.pid/etc/dhcp6.
0x00422235 hitl_35 .dhcp6s.pid/etc/dhcp6sctlkey/etc/d.
0x00422249 hitl_36 .6sctlkey/etc/dhcp6s.confusage:.
0x0042225f hitl_37 ..confusage: dhcp6s [-c configfile.
0x00423078 hitl_38 .ived %s from %sdhcp6 relay: hop=%d, .
0x004232d5 hitl_39 .p:P:%d/var/dhcp6s_duid547ff.
0x00423ba8 hitl_40 .rdvalueduidcpydhcp6_reset_timerd.
0x00423bbc hitl_41 .6_reset_timerdhcp6_set_timeoparam.
0x00423c58 hitl_42 .copy_optiondhcp6_set_addrdhcp6.
0x00423c68 hitl_43 .dhcp6_set_addrdhcp6_set_domaind.
0x00423c7c hitl_44 .6_set_domaindhcp6_set_optionsg.
0x00423cc0 hitl_45 .ddrgetifaddrdhcp6_auth_replaychec.

```

```

0x00423cd8 hitl_46 .th_replaycheckdhcp6_remove_eventd.
0x00423cec hitl_47 .6_remove_eventdhcp6_create_eventd.
0x00423d00 hitl_48 .6_create_eventdhcp6_add_listval.
0x00423d38 hitl_49 .,d,4dhcp6_get_addrdhcp6.
0x00423d48 hitl_50 .dhcp6_get_addrdhcp6_get_domainc.
0x00423d6c hitl_51 .copyin_optiondhcp6_get_optionsX.
0x00424b21 hitl_52 .e (%d)cause is dhcp6s(%d), dont send.
0x00426400 hitl_53 .dr=%s)dhcp6_add_timerdhcp6.
0x00426410 hitl_54 .dhcp6_add_timerdhcp6_timer_rest.
0x00426550 hitl_55 .=dhcp6_ctl_initdhcp6.
0x00426560 hitl_56 .dhcp6_ctl_initdhcp6_ctl_authinitd.
0x00426574 hitl_57 .6_ctl_authinitdhcp6_ctl_closecomman.
0x0042658c hitl_58 .l_closecommanddhcp6_ctl_acceptcommma.
0x004265a4 hitl_59 .l_acceptcommanddhcp6_ctl_readcommand.

```

结论：这是一个完整的 DHCPv6 服务端实现，包含选项解析、缓冲区管理、列表操作、报文发送等完整功能。

```

[0x004026d0]> pdf @ main | grep -E "call|jalr" | head -20
0x00408f04 09f82003 jalr t9
0x00408fb4 09f82003 jalr t9
0x00408fcc 09f82003 jalr t9
0x00408fe4 09f82003 jalr t9
[0x00409008] 09f82003 jalr t9
[0x00409060] 09f82003 jalr t9
[0x00409078] 09f82003 jalr t9
[0x004090bc] 09f82003 jalr t9
[0x004090e8] 09f82003 jalr t9
[0x00409100] 09f82003 jalr t9
[0x00409120] 09f82003 jalr t9
[0x00409154] 09f82003 jalr t9
[0x00409174] 09f82003 jalr t9
[0x00409198] 09f82003 jalr t9
[0x004091b0] 09f82003 jalr t9
[0x004091e4] 09f82003 jalr t9
[0x00409220] 09f82003 jalr t9
[0x00409240] 09f82003 jalr t9
[0x00409268] 09f82003 jalr t9
[0x00409294] 09f82003 jalr t9

```

结论：找到了 main 函数中大量的 jalr t9 指令，表明 main 函数通过 **20+个函数调用** 实现核心逻辑，程序结构清晰，模块化程度高。

关键函数调用一览表：

调用类型	函数名	推测功能
网络接收	recvmsg	接收客户端 DHCP 请求
选项解析	dhcp6_get_options	解析 DHCPv6 选项字段

调用类型	函数名	推测功能
选项设置	dhcp6_set_options	构造响应选项
地址管理	is_leased	检查地址租约状态
报文发送	transmit_sa	发送 DHCP 响应报文
配置解析	configure_host、configure_global	读取配置文件

3.7 控制流分析-分支指令

通过 radare2 分析 main 函数中的分支指令：

```
[0x004026d0]> pdf @ main | grep -E "beqz|bnez" | head -10
├── 0x00408f10 02004010 beqz v0, 0x408f1c
├── 0x00409028 0a004010 beqz v0, case.0x40904c.1
├── 0x004090d8 32026010 beqz v1, 0x4099a4
├── 0x0040910c be014010 beqz v0, 0x409808
├── 0x0040912c 77024014 bnez v0, 0x409b0c
├── 0x00409144 1f026010 beqz v1, 0x4099c4
├── 0x00409180 0e004010 beqz v0, 0x4091bc
├── 0x004091c4 0d004010 beqz v0, 0x4091fc
├── 0x004091d4 e4024014 bnez v0, 0x409d68
├── 0x0040922c 28024014 bnez v0, 0x409ad0
```

结论：main 函数中存在 10 个以上的条件分支指令（beqz/bnez），主要基于函数返回值（寄存器 v0/v1）进行判断。这表明程序包含：**错误处理逻辑**（函数返回非 0 值时跳转到错误处理代码）、**请求类型识别**（根据不同条件执行不同处理分支）、**配置验证**（检查配置有效性）

四、综合推断

基于以上静态分析结果，对该程序的功能做出以下综合推断：

分析维度	结论
程序名称	dhcp6s
程序类型	网络服务程序（守护进程）

分析维度	结论
核心功能	DHCPv6 服务端
目标平台	MIPS 架构嵌入式设备（路由器/交换机）
运行方式	后台守护进程，持续监听网络请求

五、总结

本次二进制分析实践以 dhcp6s 文件为分析对象，综合运用了多种静态分析技术，完成了从文件识别到功能推断的完整分析流程。

5.1 主要工作与成果

阶段	工作内容	使用工具
环境准备	搭建 WSL Ubuntu 分析环境	WSL、apt
文件识别	确认 ELF 32 位 MIPS 架构	file
完整性验证	计算并比对 SHA256 哈希值	sha256sum
导入表分析	识别网络、DHCP、文件操作等函数	readelf
字符串提取	定位配置文件、PID 文件路径	strings
反汇编分析	分析 main 函数结构与函数调用	radare2
控制流分析	识别分支指令与执行逻辑	radare2

5.2 核心结论

dhcp6s 是一个运行在 MIPS 架构嵌入式设备上的 DHCPv6 服务端程序。其主要功能包括：读取配置文件（/etc/dhcp6s.conf）、监听客户端 DHCPv6 请求、分配 IPv6 地

址、管理地址租约、记录运行日志。程序采用守护进程方式运行，通过主循环持续处理网络请求。

5.3 技术方法总结

本次分析运用了二进制分析的核心方法论：先通过 `file` 和 `readelf` 进行文件格式识别与结构分析，再用 `strings` 提取人类可读信息，最后用 `radare2` 进行反汇编和控制流分析。各工具相互配合，形成了从宏观到微观、从静态特征到动态逻辑的完整分析链条。

5.4 不足与展望

由于该程序为 MIPS 架构，本次分析未进行动态调试。后续可在 QEMU 模拟环境下运行程序，结合 `ltrace`/`strace` 跟踪系统调用和库函数调用，进一步验证静态分析结论。